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AUTONOMOUS

An AI-Powered Skills-First Recruitment and Talent Assessment Platform

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ABSTRCT

Talvix is an AI-powered skills-first recruitment and talent assessment platform designed to bridge the gap between qualified candidates and recruiters through intelligent hiring automation, transparent evaluation, and data-driven decision-making. Traditional recruitment systems primarily rely on resumes and manual screening, which often result in biased hiring, inefficient candidate evaluation, and poor skill matching. Talvix addresses these challenges by integrating artificial intelligence with a modern recruitment workflow that emphasizes practical skills, verified assessments, and recruiter productivity.

The platform provides dedicated portals for **Candidates, Recruiters, Organizations, and Administrators**, enabling end-to-end recruitment management from job posting to final hiring. Candidates can create professional profiles, apply for jobs, participate in coding challenges, MCQ examinations, hackathons, and AI-generated technical assessments. Recruiters can publish job openings, manage applications, evaluate candidate performance through comprehensive analytics, and communicate directly with applicants. Organizations benefit from centralized hiring management, while administrators oversee platform operations, user verification, and system monitoring.

Talvix incorporates an intelligent AI layer that automatically extracts required skills from job descriptions, generates customized assessments, performs semantic candidate-job matching, analyzes candidate performance, recommends career opportunities, and detects potentially fraudulent job postings. The system further supports AI-assisted interview evaluation, explainable hiring recommendations, and recruiter analytics to improve recruitment quality while reducing screening time.

The platform is developed using the **MERN Stack (MongoDB, Express.js, React, and Node.js)** with a scalable RESTful architecture. AI services are implemented using **Python FastAPI**, modern embedding models, and Large Language Models (LLMs), allowing advanced semantic analysis and intelligent automation. Additional integrations such as cloud storage, secure authentication, real-time communication, and online coding evaluation provide a production-ready recruitment ecosystem.

By combining recruitment management, automated assessments, AI-driven decision support, and intelligent candidate evaluation into a single platform, Talvix aims to transform conventional hiring into a faster, more accurate, transparent, and skills-focused recruitment process. The proposed system demonstrates how artificial intelligence can significantly improve hiring efficiency while providing equal opportunities for candidates based on demonstrated competencies rather than traditional resume-based filtering.